Choosing from a Pareto set, and being able to defend the choice

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Abstract

A multicriteria model usually ends with a set of efficient alternatives, not with a recommendation. Turning that set into one choice is normally done by a scalarisation, which needs weights, a reference point or a metric that the decision maker often cannot defend, and whose answer moves when those inputs move. This paper treats the last step as a computational problem instead. The decision maker declares a finite family of monotone scalar questions, called probes. Each alternative gets one normalised *disappointment* per probe, its excess over the best value attained on the candidate set, rescaled by the range that set actually spans. Alternatives are then compared by the classical lexicographic maximum order on their sorted disappointment vectors. We give four computational results for this construction. An exact finite procedure. A branch-and-bound that certifies, over a whole box of admissible normalisation bounds and without sampling it, which alternatives can win. A sampled mode with an explicit coverage guarantee. And a formulation that solves the same selection over a polyhedron in $O(Km)$ linear programs for K declared probes and m criteria, with no enumeration of the efficient set. We prove existence, Pareto compatibility and an exact bound-invariance property, and we show by an explicit counterexample that the rule depends on the candidate set, which is why it reports a stability class rather than a point whenever the bounds are uncertain. A study over five thousand instances then locates the source of the observed stability: it comes from the way the disappointments are built, not from the lexicographic order.

Keywords: Multiple criteria analysis, Regret minimisation, Auditable recommendation, Stability analysis, Preference elicitation

1 The problem, and why it is computational

A multicriteria model normally ends one step short of a decision. It returns the efficient alternatives, those where no criterion can improve without another one getting worse [1], and with several conflicting criteria that set can be large. Producing it is already hard work [2], but the decision maker is left holding a menu.

The usual way to close the gap is a scalarisation. Weighted sums, distance to an ideal point as in TOPSIS [3] or compromise programming [4], and achievement scalarising functions [5] all collapse the menu to one point. They work, and once their parameters are fixed they behave well. The difficulty is the parameters. A weight vector, a reference point or a metric has to be committed to, and in practice the decision maker often cannot defend the third digit of any of them. Worse, the answer moves when those inputs move: rankings under this kind of normalisation are known to be unstable when the bounds or the candidate set change [6].

The response in the literature has mostly been to model the uncertainty about preferences. Stochastic multicriteria acceptability analysis puts a distribution on the weights [7, 8]. Robust ordinal regression reasons over all value functions compatible with elicited judgments [9]. Preference robust optimisation maximises a worst-case expected utility over an ambiguity set [10, 11]. The line continues in distributional and group settings [12]. All of it answers one question: what if the preference parameters are wrong.

This paper asks a different question, and the difference is the point of the paper. Suppose the preference input is fixed and declared. There is still a set of numbers the rule depends on that nobody elicited at all: the normalisation bounds, the best and worst values used to put criteria on a common scale. Those are estimates. Once we admit that, three questions arise, and none of them is a preference-modelling question.

Which alternatives can win over an entire set of admissible bounds, rather than at one guessed value? Can that set be computed exactly, or at least enclosed with a guarantee, instead of estimated by simulation? And can a third party recompute the answer from an exported file? These are branch-and-bound, linear-programming and reproducibility questions, and they are the ones this paper answers. That places it alongside work which studies multicriteria models for their deployment [13], solves lexicographic objectives rather than only characterising them [14], and uses interval-valued dominance to report what the data can and cannot separate [15].

The rule these questions are asked about

We need a concrete selector to compute with, and we use the simplest one that supports the whole programme. The decision maker declares a finite family of monotone scalar questions, which we call *probes*: for instance "how bad is this alternative on cost", "how bad is it on average", "how bad is it on its own worst criterion". Each alternative receives one *disappointment* per probe, its excess over the best value any candidate achieves on that probe, divided by the range the candidates actually span on it. Alternatives are then compared by sorting their disappointments worst first and taking the lexicographic minimum. We call this Lexicographic Probe-Regret, LexPR.

Two things about this are deliberately unoriginal. The comparison is the classical lexicographic maximum, or leximax, order [16], and we claim nothing for it. The variational identity that makes the continuous version tractable is the conditional value-at-risk formula of Rockafellar and Uryasev [17], equivalently the linear representation of the sum of the k largest elements of Ogryczak and Tamir [18]. What is ours is the object those classical tools are applied to, and the four computational results in Section 4.

This does not remove preference modelling, and saying that it does would be false: the default family contains an equal-weight mean, which is a weight vector, and declaring a question twice is a way of emphasising it. What changes is the *form* of the input, from one vector of numbers that has to be right all at once to a multiset of named questions that can be written down, justified and disputed one by one. Figure 1 places that input in the whole procedure.

Contribution and Positioning

The novelty is not the leximax order or the top-p linearisation separately, but their integration with candidate-range re-anchored probe disappointments, certified possible-winner computation over uncertain normalisation bounds, direct optimisation without front enumeration, and an exported comparison-replay record.

This ties the paper to several literatures. It shares the mathematical machinery of equitable and ordered optimisation, but applies it to an adaptive probe profile rather than to raw attributes. It provides necessary and possible preference relations over uncertain bounds, connecting it to interval MCDA and robust ordinal regression, but without eliciting preference parameters from the decision maker. It ensures parametric robustness of the normalisation bounds, but tests it structurally rather than via scalarised sampling.

Four components carry this, all in Section 4. An exact finite procedure. A branch-and-bound that certifies, over a whole box of admissible normalisation bounds and without sampling it, which alternatives can win. A sampled mode with an explicit pointwise coverage guarantee, for when certification is out of budget. And a formulation that solves the same selection over a polyhedron

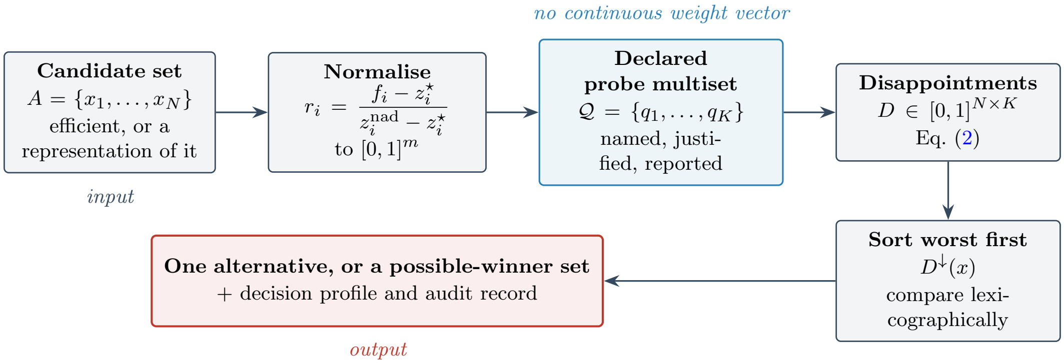

Fig. 1: The whole method on one line. The candidate set and the declared probe family go in. A recommendation, or an explicit set of possible winners that cannot be separated, comes out, together with the reasons. The list of candidates must be frozen before selection.

with affine criteria in $O(Km)$ linear or mixed-integer linear programs. Section 3 establishes what the rule itself guarantees, and Section 5 reports what five thousand instances say about it, including one result that runs against the method.

Table 1 states at the outset what is proved, what is only observed, and what is not claimed. It is given once and not restated.

The rest of the paper follows the order in which someone would actually use the method. Section 2 builds the rule and shows it working on a three-alternative example small enough to check by hand. Section 3 says what it guarantees and, just as importantly, where it breaks. Section 4 computes it. Section 5 tests it. Section 6 says when to use it. Everything omitted here, including the full proofs, the algorithm listings and the raw outputs, is in the reproduction archive [19], version 1.0.4.

2 The selector

2.1 From criteria to disappointments

Let $F : X \to \mathbb{R}^m$ be the vector of criteria, all minimised, and let $A = \{x_1, \ldots, x_N\} \subseteq X$ be the candidate set. Throughout, A is finite and nonempty, which is part of the model contract: on an empty set the order below is vacuously complete but no winner exists. Criteria are put on a common scale by

$$r_i(x) = \frac{f_i(x) - z_i^\star}{z_i^{\text{nad}} - z_i^\star}, \tag{1}$$

with $z_i^\star = \min_{x \in A} f_i(x)$ and $z_i^{\text{nad}} = \max_{x \in A} f_i(x)$. Both endpoints are attained on A, so every r_i lands in $[0,1]$. A criterion whose relative range on the scale falls below a threshold ε_z carries no information and is excluded from aggregate probes. We use $\varepsilon_z = 10^{-9}$ everywhere.

Two conventions are possible here and they are not equivalent. Computing the bounds on A ties the scale to values the decision maker can verify on the shortlist in front of them. Fixing them externally makes the rule insensitive to shortlist edits. We use active-set bounds because they are verifiable, and Section 3 is honest about what that costs. Whichever is chosen must be declared.

A *probe* is a scalar question about a normalised alternative.

Definition 1 (Admissible probe). Let $\mathcal{R} \subseteq \mathbb{R}^m$ be a domain containing every $r(x; b)$ generated by the candidate set and any admissible bound vector. A function $q : \mathcal{R} \to \mathbb{R}$ is an admissible probe when it is non-decreasing in every r_i over $\mathcal{R}$, and can be rescaled by a positive affine map so that $q(\mathbf{0}) = 0$ and $q(\mathbf{1}) = 1$ under the nominal bounds.

Monotonicity says the probe agrees with the direction of the criteria over the entire reachable domain. The normalisation excludes probes that are constant. The declared family $\mathcal{Q}$ is a *multiset*: a question may be declared twice, and that repetition is a strong preference parameter, changing the order after ties in the leading coordinates.

Table 1: What this paper claims. "Proved" means proved under the model contract of Section 2. ASF denotes achievement scalarisation and MMR minimax regret. A subscript K marks the variant applied to the same probe disappointments as LexPR rather than to the raw normalised criteria.

Claim	Status	Conditional on
Existence, completeness, Pareto compatibility	Proved	finite nonempty A, retained family separates criteria
Exact bound-invariance of single-criterion probes	Proved	retained singleton probes only
Soundness of the certified inner and outer sets	Proved	canonical probes, terminal-leaf aggregation
Coverage of the sampled stability mode	Proved	declared sampling distribution
Correctness of the polyhedral formulation	Proved	affine criteria, polyhedral X, canonical probes
Audit record rebuilds the winner class	Proved, and checked by an independent verifier	complete record, comparison replay
Invariance under interior candidate-set edits	Proved	frozen family, no anchor moves
Unrestricted independence under candidate-set-dependent normalisation	**False**, by explicit reversal witness	active-set anchoring
Redundancy control by cluster probes	**Not established**, reported as a diagnostic only	—
Lower point fragility than raw-normalised ASF	Supported, but caused by re-anchoring and not by the leximax	generator family, Section 5
Lower fragility than a scalarisation on the *same* probes	**False**: ASF_K matches or slightly beats LexPR	Section 5
Lower held-out loss than ASF	**Not established**: the family-clustered interval contains zero	Section 5
General superiority over rival rules	**Not claimed**	—

For each probe let $q^\star = \min_{x\in A} q(r(x))$ and $q^{\mathrm{w}} = \max_{x\in A} q(r(x))$ be the best and worst values it takes on the candidate set, and let $R_q = q^{\mathrm{w}} - q^\star$ be its *active range*. When $R_q > 0$ the *disappointment* of x under q is

$$D_q(x) = \frac{q(r(x)) - q^\star}{R_q} \in [0,1]. \tag{2}$$

Read it plainly: zero means "as good as anything on the table on this question", one means "the worst on the table". Probes with $R_q \le \varepsilon_z$ are dropped for that run and reported as uninformative, because keeping them would pad every profile with a zero and misstate the family size.

The family we use throughout is the *canonical core*

$$\mathcal{Q}_{\mathrm{can}} = \{q_1,\dots,q_m\} \cup \{q_\mu, q_{\max}\}, \qquad K = m+2, \tag{3}$$

made of the m single-criterion probes $q_i(r) = r_i$, the mean $q_\mu(r) = m^{-1}\sum_i r_i$, and the maximum $q_{\max}(r) = \max_i r_i$. In words: one question per criterion, plus a utilitarian reading of the whole and a protective one. The single-criterion probes are what let the output name a criterion, and the mean and the maximum are what let it express a compromise attitude. It is the smallest family containing those three readings, in this precise sense: if $m \ge 2$ and neither aggregate is a positive affine transform of a singleton or of the other aggregate on the active range, no smaller family provides all three. The counting argument is in

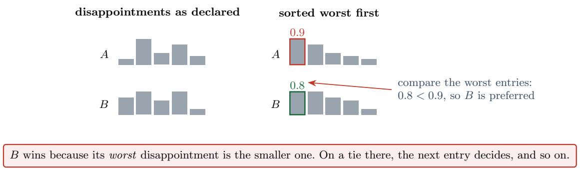

Fig. 2: The comparison, once. Each alternative's disappointments are sorted worst first and the two sorted vectors are read left to right until they differ.

the archive. We do not claim that $m+2$ probes are forced by Pareto compatibility alone, nor that the arithmetic mean is the only reasonable compromise aggregate: other declared readings, such as the mean over a regulated subset, enlarge the family accordingly. What the core does guarantee is that K stays linear in m.

Real criteria sometimes overlap, two of them partly measuring the same thing. We report that as a diagnostic and do not compensate for it inside the family. An optional layer adds one mean and one maximum probe per correlated cluster, and the archive implements it, but it is not recommended and was empirically inert in everything reported here. Because all single-criterion probes are kept, it cannot remove approximate duplicates, and calling it redundancy control would overstate it.

2.2 Comparing alternatives

Collect the disappointments of x into $D(x) \in [0,1]^K$ and sort them in decreasing order to get $D^{\downarrow}(x)$. Then

$$x \preceq_{\mathrm{LexPR}} y \iff D^{\downarrow}(x) \leq_{\mathrm{lex}} D^{\downarrow}(y). \qquad (4)$$

The comparison starts at the worst entry of each sorted vector and moves right only on an exact tie. Figure 2 shows it once, on two alternatives, and the reader who follows that figure has understood the rule.

An optional tolerance $\tau \geq 0$ can absorb estimation noise by treating coordinates that differ by less than τ as tied. We mention it only to recommend against it. For $\tau > 0$ the relation is not transitive and its undominated set can be empty, so the implementation falls back to the exact class with a warning whenever the comparison graph contains a cycle. At $\tau = 0$, which is the recommended setting and the one used in every number in this paper, the relation is a strict weak order and the induced order on equivalence classes is total, and the whole layer disappears.

2.3 What comes out, and why it can be checked

Three objects are reported alongside the winner, and they do different jobs.

The *decision profile* is the list of labelled disappointments, $\big((q, D_q(x))\big)_{q\in\mathcal{Q}}$, sorted worst first. It is for interpretation. It says which declared questions the recommendation is worst on, and by how much. It does not prove optimality, and because sorting discards the original labels before comparing, the coordinate deciding a pairwise comparison need not correspond to the same probe for both alternatives. Because the recommendation is determined by exactly this sorted vector, the explanation is computed from the data that drive the selection rather than reconstructed afterwards.

The *contrastive record* does prove optimality, relative to the profiles. For a unique winner $x^\star$ and each rival y, it stores the first sorted coordinate $k(y)$ at which $x^\star$ is strictly lower and the margin $D^{\downarrow}_{k(y)}(y) - D^{\downarrow}_{k(y)}(x^\star) > 0$. The smallest such margin, $\Delta_{\mathcal{C}}(x^\star)$, measures how close the decision was. It will return in Section 5 as a cheap predictor of whether the answer will be a point or a class.

The *audit record* is the complete file: the criterion matrix and its hash, the bounds, the declared multiset with versions and multiplicities, the anchors, the degeneracy threshold, which probes were retained and which dropped, every candidate's profile, the comparison rule, the winner class and the margins. An independent program reading only this file rebuilds the winner class and the margins, and the archive ships one that does. The labelled disappointments alone suffice, since the comparison never reads the anchors, the bounds or the probe definitions. Reporting only the binding probes is not enough: the order cannot then be rebuilt in general.

One term used throughout deserves a definition here. When the bounds are uncertain the output is not one alternative but a *possible-winner set*: the set of alternatives that win under at least one admissible bound vector, written S^{pos} in

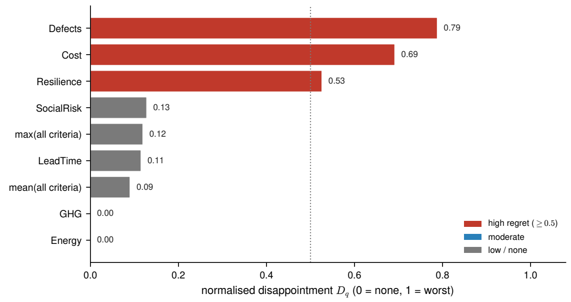

Fig. 3: A decision profile as delivered. Bars above the reporting threshold are the questions on which the recommendation is weakest and are drawn darker. The two near-identical environmental criteria both carry zero disappointment. The figure reads in greyscale, since shade repeats position.

Section 4 and approximated there either by sampling or with a guarantee. “Winner class” means something narrower, the set of alternatives tied under one fixed bound vector.

One qualification is easy to get wrong, so we fix it once. Replay comes in three levels. *Comparison replay* takes the exported disappointments and re-derives the order, the winner class and the margins. It checks the reasoning, not the arithmetic that produced its inputs. *Probe-evaluation replay* additionally recomputes the disappointments from the criterion matrix, the bounds and the probe definitions, which checks the anchors and the normalisation too. *Model replay* additionally regenerates the candidate set from the underlying optimisation model. The shipped verifier does comparison replay. Whenever this paper says a result is independently verified, that is what it means.

Figure 3 shows a real profile, from the supplier case of Section 5. It is the whole output of the method on one instance: which questions the recommended supplier is uncomfortable on, and which it answers with room to spare.

3 What the rule guarantees, and where it breaks

This section states the properties of the construction. Each is given with the idea of its proof, with complete derivations in the archive [19]. The order runs from the reassuring to the uncomfortable, and the uncomfortable result is the one that shapes how the method is used.

The rule always answers

For finite nonempty A and finite $\mathcal{Q}$, a LexPR-optimal alternative exists and $\preceq_{\text{LexPR}}$ is a complete preorder on A. The argument is one line: the lexicographic order is total on $\mathbb{R}^K$, hence total on the finite image $\{D^{\downarrow}(x)\}$, and a total order on a finite nonempty set has a least element. Reflexivity and transitivity are inherited. Ties in $D^{\downarrow}$ are what make it a preorder rather than an order, and they are reported as ties rather than broken silently. A deployment therefore needs no fallback for the case of no winner.

It never picks a dominated alternative

Suppose the bounds and anchors are those of the fixed set A, the family $\mathcal{Q}$ is the one retained after dropping degenerate probes, and every retained probe is non-decreasing in each r_i. If x dominates y in the Pareto sense, then $D_q(x) \le D_q(y)$ for every q by monotonicity, so sorting preserves the inequality coordinatewise and $x \preceq_{\text{LexPR}} y$. If moreover the retained family *separates criteria*, meaning that for each criterion some retained probe is strictly increasing in it, the inequality is strict somewhere and $x \prec_{\text{LexPR}} y$. This is a safety property, not a claim that the rule recovers anyone’s true preference. The qualifier “retained” is not cosmetic: if degeneracy removal deletes the only probe that was strictly increasing in the criterion on which x beats y, the conclusion can fail. The canonical core contains all m single-criterion probes, so it separates by construction.

Single-criterion disappointments do not depend on the bounds at all

This is the least obvious property and the most consequential.

Proposition 1 (Exact bound-invariance)**.** *Let $q_i(r) = r_i$ be a retained single-criterion probe, so criterion i has positive active range on A, and let $(z_i^\star, z_i^{\text{nad}})$ be* any *pair with $z_i^{\text{nad}} > z_i^\star$, not necessarily the active one. Then*

$$D_{q_i}(x) = \frac{f_i(x) - \min_{y\in A} f_i(y)}{\max_{y\in A} f_i(y) - \min_{y\in A} f_i(y)}, \qquad (5)$$

which does not involve $(z_i^\star, z_i^{\text{nad}})$.

Proof idea. Write $R_i = z_i^{\text{nad}} - z_i^\star > 0$. Then $r_i(x) = (f_i(x) - z_i^\star)/R_i$ is a positive affine function of $f_i(x)$, and the anchors of q_i over A are the same affine function of $\min_y f_i(y)$ and $\max_y f_i(y)$. Substituting into (2), the offset $-z_i^\star/R_i$ cancels between numerator and denominator and the factor $1/R_i$ cancels as well. □

The consequence is sharp. A family made only of retained single-criterion probes induces a selection that is *exactly* invariant to the declared bounds. Only aggregate probes, which mix criteria whose ranges move at different rates, carry any bound dependence. Section 5 measures this and finds precisely what (5) predicts. The practical reading is narrow, though: invariance holds for singletons *alone*, and one active aggregate probe is enough to lose it.

Repeating every question is safe, repeating one is not

Replicating every probe the same number of times leaves the order unchanged, because the sorted vector is repeated blockwise and lexicographic comparison of blockwise repetitions agrees with the original. If two alternatives differ at their worst disappointment, duplicating one probe cannot change their comparison either, since the worst entry stays first after sorting. Once the worst entries tie, selective replication can reverse a comparison, and the archive holds a three-probe instance where one duplicated question flips the recommendation. That is why multiplicities must be declared and why the deduplication check is part of the protocol, not an optional extra.

The rule depends on the candidate set, and that cannot be removed

Active-set normalisation computes both the criterion bounds and the probe anchors from A. Editing A can therefore move those anchors, and moving them changes the rule itself, not merely its input.

Theorem 1 (Failure of unrestricted independence)**.** *There exist finite sets $A \subset A'$ and alternatives $x, y \in A$ such that active-set LexPR ranks y before x on A and x before y on A', when bounds and anchors are recomputed on each set.*

Proof idea. A four-point witness suffices. On $A = \{x, y, z\} \subset \mathbb{R}^2$ the active box is spanned by the three points and y beats x. Adding $w = (4, 11)$, which dominates neither, stretches the criterion-1 ideal and the criterion-2 nadir. Every disappointment is re-anchored, and the comparison reverses. The numerical values are machine-checked in the archive. □

We are deliberate about how to name this. "Independence of irrelevant alternatives" means different things in social choice, discrete choice and multicriteria analysis, and Theorem 1 does not address all of them. What it establishes is the failure of *unrestricted independence under candidate-set-dependent normalisation*, and what it exhibits is a rank reversal of that kind. Every use of "IIA" below is shorthand for exactly that.

This is not a defect peculiar to LexPR. Every rule that normalises on the observed set inherits it, as the supplier experiment of Section 5 confirms by measuring the same effect on three standard baselines. What can be done is to say precisely when it does not bite.

Theorem 2 (Invariance under interior edits)**.** *Let A be finite with retained family $\mathcal{Q}$ and a unique winner $x^\star$. Form A' by adding an alternative z or removing a non-winner y, and suppose the edit leaves the retained family and its degeneracy statuses unchanged, and moves no active extremum. Concretely, in the addition case $z_i^\star \le f_i(z) \le z_i^{\text{nad}}$ for every criterion and $q^\star \le q(r(z)) \le q^{\text{w}}$ for every retained probe, and in the removal case y is not the unique attainer of any criterion or probe extremum. Then every D_q is unchanged on $A \cap A'$, so the order there is preserved, and $x^\star$ still wins A' provided the added z does not beat it, which holds whenever z is dominated and $\mathcal{Q}$ separates criteria.*

Proof idea. Under the stated conditions none of $z_i^\star, z_i^{\text{nad}}, q^\star, q^{\text{w}}$ changes value, and D_q is a function of the alternative and those four quantities alone. □

The practical content is a rule for shortlist hygiene: edits that move no extremum are safe, and edits that move one require re-opening the declared model. This also delimits what the interval analysis of the next section can see. A box over criterion bounds captures the reversals whose anchor configurations that box induces, but not an interior addition that moves a probe anchor while leaving every criterion bound fixed, which the nonlinear probe $q_{\max}$ makes possible. Full invariance would need bounds and anchors fixed externally, which is a different convention rather than a repair.

When a point recommendation survives measurement error

Let $x^\star$ be the unique winner with contrastive margin $\Delta_{\mathcal{C}}(x^\star) > 0$, and let η bound the error in the computed disappointments. Sorting is 1-Lipschitz in the supremum norm, so each sorted coordinate moves by at most η and each pairwise gap by at most 2η. If $2\eta < \Delta_{\mathcal{C}}(x^\star)$ and every rival is already separated at the first sorted coordinate, the winner is unchanged. Applying the same bound to the per-probe slacks says when the reported *explanation* is stable, not merely the winner: if 2η falls below the distance of every slack from the reporting band, the set of probes shown as binding is unchanged too. That is the stricter and more relevant requirement when a profile is put in front of a decision maker. These conditions are sufficient, not necessary, and they do not cover exact ties before the deciding coordinate, where lexicographic comparison is not uniformly stable. That gap is exactly why Section 4 reports a class rather than a point whenever the bounds are uncertain.

4 Computing the recommendation

Four computational problems follow, treated in the order a user meets them. Compute the answer on the shortlist as given. Then ask what it would be if the normalisation bounds were different, first by sampling and then, if the budget allows, with a guarantee holding over a whole continuum of bounds. Finally, when the alternatives are not a list but the solutions of a model, compute the same selection without generating that list.

4.1 The finite problem

The exact procedure fixes the active set, computes the bounds and anchors, removes degenerate probes while warning about any criterion that loses its last separating probe, evaluates the disappointment matrix, sorts each row worst first, and returns the lexicographically minimal class. Its cost is dominated by evaluating the family, not by comparing alternatives. Building the matrix costs $O\big(N\sum_q T_q\big)$ where T_q is the cost of one probe, and sorting the rows costs $O(NK\log K)$. For the canonical core, all $K = m+2$ probes can be evaluated for one alternative in $O(m)$ operations combined, and sorting them costs $O(m\log m)$. Thus, the whole selection is $O(Nm\log m)$. In the reference implementation (Python 3.11, Apple M3 Max, macOS 14.5), this resolves a hundred thousand candidates on ten criteria in a median of 200 milliseconds (IQR: 3 ms), single threaded, over a hundred repetitions. The measured interval covers normalisation, probe construction and evaluation, anchor computation, sorting and comparison. It excludes generating and dominance-filtering the candidate set, which happens upstream. The number matters not because selection is a bottleneck, but because the analyses below re-run this procedure thousands of times.

4.2 Uncertain bounds: sampling with a guarantee

Write $\mathcal{B}$ for a box of admissible bound vectors and $W_0(b)$ for the exact winner class under bounds b. The two objects of interest are the alternatives that win under *some* admissible bound and those that win under *all* of them,

$$S^{\text{pos}} = \bigcup_{b\in\mathcal{B}} W_0(b), \qquad S^{\text{nec}} = \bigcap_{b\in\mathcal{B}} W_0(b). \tag{6}$$

Every experiment here uses one model for $\mathcal{B}$, set by a single level $p \geq 0$ and applied multiplicatively to each nominal bound, independently across criteria. The ideal $z_i^\star$ is multiplied by $1-u_i'$ with $u_i' \sim \mathrm{Unif}(0,p)$ and the nadir z_i^{nad} by $1+u_i$ with $u_i \sim \mathrm{Unif}(-p,p)$, the nadir floored above the ideal so every draw has positive range. Because perturbed nadir bounds can fall within the observed criterion range, they act as scale parameters rather than strict extremes. If strict bracketing is required, the perturbation can be constrained to the domain bounding the observed candidates. The asymmetry is deliberate: for a cost criterion the true best attainable value can lie below the observed one, but the observed one is attained. The certified mode of the next subsection takes the deterministic hull of this support, the box $\mathcal{B}(p)$. A "five per cent box" means $p = 0.05$ in this sense, not a fraction of the criterion range. Two scope conditions come with the multiplicative form. It presumes bounds of constant sign, which holds here because every criterion is a positive cost or quantity. And the sampling distribution is part of the declared model, not a neutral default, so every frequency reported under it is conditional on it.

Sampling M draws gives a lower approximation of S^{pos} with a one-sided guarantee. This is a pointwise guarantee: for any fixed alternative whose winning region has probability at least α, the probability that M independent draws all miss it is at most $(1-\alpha)^M$. By the union bound, to simultaneously observe all such alternatives with confidence at least $1-\delta$, a sample size of $M \geq \log(N/\delta)/(-\log(1-\alpha))$ is sufficient. That turns "we sampled a thousand draws" into a statement with a failure probability attached. It says nothing about winning regions of small probability, and nothing at all about regions of probability zero, which is the gap the next subsection closes. The direction of the guarantee also has to be respected. The sampled union is an *inner* approximation of S^{pos}, so finding an alternative there proves it can win. The sampled intersection is an *outer* approximation of S^{nec}, so it can only be too large, and it must never be read as a certificate that an alternative wins under every admissible bound.

4.3 Uncertain bounds: certification without sampling

The certified mode replaces sampling by interval arithmetic and branch and bound. Its starting point is Proposition 1: single-criterion disappointments do not depend on the bounds, so only aggregate probes have to be enclosed at all, and the problem is much smaller than its $2m$ nominal dimensions suggest.

Fix a sub-box $\beta \subseteq \mathcal{B}$. For fixed x and i, the normalised value $r_i(x;b)$ is linear-fractional in the two bound variables with a denominator bounded away from zero, so its extremes over the rectangle β occur at vertices and can be evaluated exactly. Each canonical probe is monotone in r, so an enclosure of $q(r(x;b))$ follows coordinatewise, and for the maximum probe it is not merely sound but exact, because the coordinate blocks are independent under a Cartesian box. The disappointment is a ratio of three quantities all driven by the same b, and the enclosure ignores that shared dependence, which makes it sound but not tight. Every operation is rounded outward, one step of the machine's next-representable-value function per elementary operation, so the enclosures are valid in floating point and not only in exact arithmetic. If the enclosed active range of a probe fails to be positive, the trivial enclosure $[0,1]$ is used and the box is subdivided rather than resolved.

Proposition 2 (Soundness of the certified test)**.** *Write $D^L(\cdot;\beta)$ and $D^U(\cdot;\beta)$ for the coordinatewise lower and upper enclosures over β. If $D^{\downarrow}\big(D^U(x;\beta)\big) \leq_{\mathrm{lex}} D^{\downarrow}\big(D^L(y;\beta)\big)$ strictly, then $x \prec_{\mathrm{LexPR}} y$ for every $b \in \beta$.*

Proof idea. Coordinatewise, $D(x) \leq D^U(x)$ and $D(y) \geq D^L(y)$. Sorting is monotone with respect to the coordinatewise order, and the lexicographic order on sorted vectors is monotone in the same sense, so the assumed strict inequality between the enclosures propagates to the true profiles. □

A negative answer from this test is therefore trustworthy in a way no sample can be: an eliminated candidate provably cannot win anywhere in the box. The search maintains a stack of sub-boxes, each carrying the set of candidates its parent could not eliminate. On each box it eliminates every candidate that some rival beats throughout, and it declares the box *resolved* when one candidate beats every rival throughout. A resolved box contributes its sole winner to the inner set and to the outer set. An unresolved box, whether depth-capped or left on the stack when the budget runs out, contributes its inherited survivor set to the outer set and its volume to the unresolved total.

Two details decide whether this is correct, and both are easy to get wrong. The outer set is accumulated over terminal leaves by *union*, never by deleting from a running global set. A candidate eliminated on one sub-box may win on another, so if a wins throughout β_1 and b throughout β_2 the possible set is $\{a,b\}$, whereas subtracting each local elimination in turn would remove first b and then a and return the empty set. And resolution uses the direct test, that some $x^\star$ beats every rival, rather than the weaker observation that $x^\star$ has no rival certified to beat it, which would need a separate acyclicity argument. Because elimination is monotone under subdivision, a parent's survivor set is sound for any descendant, including one never processed, which is what makes budget exhaustion safe. The archive contains a regression test that fails if the union is replaced by subtraction.

The procedure returns $\mathrm{In} \subseteq S^{\mathrm{pos}} \subseteq \mathrm{Out}$ and an unresolved volume fraction u. When $u = 0$ and every resolved box has the same sole winner, that winner is additionally certified necessary. This test detects a necessary winner but not a necessary *class*, so the output is a certified subset of S^{nec} rather than the set itself unless witnesses supply the rest. Two convergence caveats are permanent rather than budgetary. Exact recovery fails on persistent tie surfaces, where no sole winner exists to certify. And the inner set can miss an alternative whose winning region has empty interior, since such an alternative is never the sole winner on a box of positive volume. Its membership has to be established by an explicit witness, that is by exhibiting one admissible bound vector at which it wins.

Two radii, not one

Bisecting the level p under the same test gives a lower bound on the stability radius, and the paper keeps two quantities apart. The *true radius* of the nominal winner is $\rho^\star = \sup\{p \geq 0 : W_0(b) = \{x^\star\}\ \forall b \in \mathcal{B}(p)\}$, a property of the data. The *certification frontier* $\underline{\rho}_{B,d}$ is the largest p at which the procedure proves uniqueness under box budget B and depth cap d, and it depends on the enclosures and the budget as well as on the data. A witnessed reversal, an explicit b at which some rival wins, gives an upper bound $\overline{\rho}$. Soundness gives

$$\underline{\rho}_{B,d} \ \leq\ \rho^\star \ \leq\ \overline{\rho}, \tag{7}$$

and both inequalities can be strict. Failure to certify above the frontier may mean loose enclosures or an exhausted budget rather than genuine instability, so the frontier is never reported as the radius. Raising the budget can only raise it, and no finite budget is guaranteed to close the bracket.

Does it actually work?

Soundness is a theorem, but a sound procedure whose enclosures are too loose to prove anything would still be useless, so we test it under a dense-grid stress test. On low-dimensional instances the box has only $2m$ coordinates and can be swept exhaustively: we evaluate the exact rule at every point of a dense product grid over $\mathcal{B}(p)$ and collect the winners it produces as a set S_{grid}. Every grid point is an admissible bound vector, so $S_{\mathrm{grid}} \subseteq S^{\mathrm{pos}}$ by construction, and the inclusion $\mathrm{In} \subseteq S_{\mathrm{grid}} \subseteq \mathrm{Out}$ is a real test of the outer set: an alternative that wins somewhere on the grid but is missing from Out would be a soundness violation.

Table 2: Dense-grid stress test of the certified outer set, eleven instances at three levels. S_{grid} is the winner set found by sweeping the bound box exhaustively. The last column is the smallest budget on the ladder $250, 1{,}000, 4{,}000, 16{,}000$ at which the inner and outer sets coincide with zero unresolved volume, and is blank where no rung achieved it. Full per-row output is in the archive.

Behaviour	rows	viol.	grid ties	B^{ex}
Exact recovery	3	0	0%	250–4,000
Dependency slack	24	0	0%	—
Persistent tie	3	0	100%	—
Near-degenerate	3	0	0.5–0.8%	—
All	**33**	**0**		

Table 2 reports the outcome over eleven instances at three levels. The inclusion holds in all thirty-three cases with no violation, and monotone tightening holds everywhere: on every row the outer set and the unresolved fraction are non-increasing in the budget, as terminal-leaf aggregation requires. Three rows reach exact recovery, the unresolved volume falling to zero with the inner and outer sets coinciding as a singleton, at budgets of 250, 250 and 4,000 boxes. The rest show the failure modes rather than hiding them. A constructed instance with two identical criterion rows ties at every grid point, so no sole winner exists anywhere and the inner set stays empty: correct behaviour, not failure. A near-degenerate instance whose second criterion spans a range of 0.003 triggers the denominator fallback on 81 to 83 of the 128 pairs probed, that is 64 sub-boxes times the two aggregate probes. And on most rows the grid shows two or three winners while the outer set is still the whole candidate set, the visible cost of ignoring the shared dependence in the enclosure.

Two limits are worth naming. The grid is dense but finite, 23 points per coordinate at $m = 2$ and 8 at $m = 3$, so a winning region smaller than the grid step would be invisible to it. And the certified mode reaches only small boxes and modest criterion counts, and elsewhere the honest output is the bracket with its unresolved volume attached.

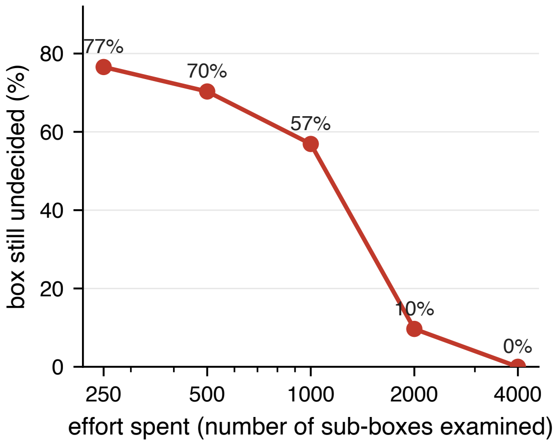

Fig. 4: Spending more effort shrinks the part of the bound box left undecided. What grows with the budget is not the validity of the answer, which holds throughout, but how much of the box the procedure manages to settle.

What certification costs

Two things govern the practical reach and they are not the same. The box count is the visible budget, but the cost per box grows with the candidate set, because certifying a box means showing that every candidate is beaten throughout. Measured cost per box rises from 0.87 milliseconds at $N = 20$ to 8.11 at $N = 100$, nearly a factor of ten for a fivefold increase in N. Pure quadratic growth would give twenty-five, so the realised cost sits between linear and quadratic, because the elimination test stops at the first rival that dominates. A hundred-alternative shortlist is *processed* in about eight seconds at a thousand-box budget, which is not the same as certified: that run left 93% of the box unresolved and returned the whole shortlist as its outer set. Time is cheap here and resolution is not, which is the point of the previous paragraph. The validated interval arithmetic costs roughly a fifth of the running time and changes no certified set, box count or unresolved fraction: switching from plain floating point to outward-rounded operations reproduces every certified number here to the digit.

Figure 4 shows how the unresolved volume falls as the budget grows, and the shape of that curve is the practical message. Quadrupling the budget from 250 to 1,000 sub-boxes takes the unresolved volume from 0.766 to 0.569, and doubling again to 2,000 takes it to 0.097. At a nominal budget of 4,000 the procedure terminates inside its budget, using 2,277 boxes and leaving no unresolved leaf. That last step is where the guarantee stops being merely sound and becomes sharp: throughout the under-resolved regime the outer set is the full candidate set, so the procedure proves nothing about who wins, only that its answer is not wrong, and the moment the unresolved volume reaches zero the outer set collapses from fifteen alternatives to one. Soundness holds at every budget, sharpness arrives all at once. The unresolved fraction, not the running time, is the number that tells an analyst which regime a run is in.

4.4 When the alternatives are not a list

Sometimes the candidate set is not given but implied by a model, and enumerating it is the expensive part. If the feasible set X is a nonempty bounded polyhedron, the criteria are affine, and the probes are weighted sums or maxima, the same selection can be computed directly.

The tool is the variational identity of Rockafellar and Uryasev [17] and Ogryczak and Tamir [18]: the sum of the p largest of a finite collection $d_1, \dots, d_K$ equals $\min_t \big(pt + \sum_q \max\{d_q - t, 0\}\big)$. Lexicographic minimisation of the sorted vector is exactly the sequential programme that first minimises the largest disappointment, then among its minimisers the sum of the two largest, and so on. For $p = 1, \dots, K$, stage p solves:

$$\begin{aligned} \min_{x,t,e} \quad & pt + \sum_{q \in \mathcal{Q}} e_q \\ \text{s.t.} \quad & e_q \geq D_q(x) - t, \quad e_q \geq 0 \quad \forall q \in \mathcal{Q} \\ & x \in X_p, \end{aligned}$$

where $X_1 = X$ and X_{p+1} restricts X_p by imposing the optimal objective value of stage p. Since

$D_q(x)$ is affine for weighted sums, and the maximum probe $q_{\max}$ can be represented by a standard epigraph variable, each stage becomes a linear programme.

Proposition 3 (Solver call count)**.** *For the canonical core, the exact procedure requires exactly* $6m + 5$ *linear programmes.*

Proof. Finding the criterion bounds requires $2m$ programmes. The m single-criterion probes and the mean probe constitute $m + 1$ weighted-sum probes, requiring $2(m + 1)$ programmes for their anchors. The lower anchor of the maximum probe is a single minimax programme, and its upper anchor requires m maximisation programmes, totalling $m + 1$. Finally, the $K = m + 2$ stages require one programme each. Summing these yields $2m+2(m+1)+m+1+(m+2) = 6m+5$. Integrality in X turns each linear programme into a mixed-integer one without changing the count. □

Theorem 3 (Polyhedral computation)**.** *Under affine criteria, a nonempty bounded polyhedral* X *and canonical probes, the sequential procedure computes the lexicographically minimal sorted profile in a finite sequence of linear programmes, or mixed-integer linear programmes under integrality, and characterises the optimal solution set through the final stage-optimality constraints.*

Proof idea. Correctness of the sequence is the standard equivalence between leximax minimisation and the nested top-p programme. Finiteness and attainment follow from boundedness. □

Three qualifications belong with this result and are usually left out. First, the optimal set may be infinite, and it need not be a face of X: minimising a convex piecewise-linear function over a polytope can select a relative interior point, as $\min\max\{x, 1 - x\}$ over $[0, 1]$ does at $x = 1/2$. The procedure therefore characterises a polyhedral optimal subset, and a declared secondary rule, itself recorded, picks a representative when one point is required.

Second, the word "exactly" holds under exact optimisation. A floating-point solver returns optima up to its tolerance, and imposing a previously reported optimum as a hard constraint on a later stage is not safe: the stage can become infeasible, or can exclude points that were optimal in exact arithmetic. The implementation therefore imposes the carried optimum inflated by $\epsilon_{p'} = \texttt{tol} \cdot (1 + |T^{\star}_{p'}|)$ with $\texttt{tol} = 10^{-6}$, recorded in the audit record with the solver identity. The inflation is relative by design, because stage optima grow with the stage, being sums of order statistics, so a fixed absolute tolerance that is adequate at the first stage is too tight at the last and shows up as a spurious infeasibility. The price is that the recovered set is optimal for an ϵ-relaxed sequence and can be strictly larger than the exact one. This is a solver-tolerance matter, unrelated to the interval certification above.

Third, the anchors here are computed over X, whereas the finite procedure computes them over the filtered candidate set. Those anchors differ, so the two versions need not agree. Avoiding enumeration of the efficient set is not the same as reproducing the answer one would get from it, and which convention is intended has to be declared.

What the computational study does and does not show

The scheme is implemented in full and measured against a baseline that draws 400 weight vectors, solves a weighted-sum programme for each, and applies the leximax comparison to the resulting points. Both rules use Gurobi 11.0.2. Figure 5 reports the cost over fifty instances. The realised solver-call count is $6m + 5$ at every size, and it matches the scheme term by term rather than merely respecting its order. The direct formulation uses about an order of magnitude fewer solver calls, 53 against 443 at $m = 8$, and runs five times faster. The baseline budget is held fixed at four hundred weight vectors at every size, so the comparison does not test how that budget would have to grow with m to keep the sampled set representative. The claim is that the direct cost is linear in m against a fixed baseline budget, not that the gap provably widens.

We do not present the accompanying quality comparison as validation of optimality, and it would be misleading to do so. In all fifty instances the direct solution was lexicographically no worse than the best sampled point, but that is close to tautological. The baseline searches a finite subset of the same feasible set, and weighted sums cannot reach unsupported efficient points at all,

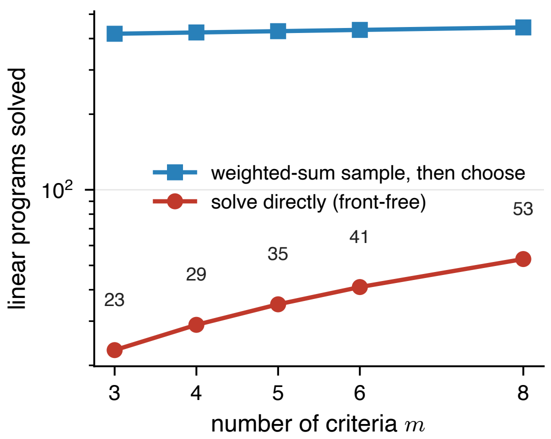

Fig. 5: Solving the selection directly over the polytope costs a small and predictable number of linear programmes, exactly $6m + 5$ for the canonical family, while weighted-sum sampling costs hundreds whatever the criterion count. The vertical scale is logarithmic.

so the direct solution must be no worse. Calling four hundred weighted sums an enumeration of the efficient set would be wrong. Two gaps are open rather than hidden: an independent check against exact vertex enumeration on small problems would test optimality properly and is not attempted, and Theorem 3 covers mixed-integer sets while the study exercises polytopes only, so that branch is proved and not benchmarked.

5 What the experiments show

The question this section answers is not whether LexPR beats other rules. It is which part of the construction produces the behaviour we observe. The answer, stated first so that no figure below can be read as supporting a different one: the observed reduction in point fragility comes from the disappointment construction, and specifically from re-anchoring each probe on its own active range. The lexicographic order contributes nothing measurable to it. What the lexicographic order contributes is the labelled worst-first explanation.

We evaluate 500 candidate sets from six generators: knapsack and job-shop instances from combinatorial and ε-constraint procedures, WFG2 [20] and DTLZ [21] sampled by Latin hypercube [22], and Energy [23] and Concrete [24] surrogate matrices. Sets reach $N = 500$ points and $m = 15$ criteria, and none is an exactly enumerated Pareto front, which is why we call them candidate sets throughout. The declared family is the canonical core (3) unless stated otherwise, and no instance produced a correlated cluster at the declared threshold of 0.6, so the optional layer coincides with the core everywhere here.

Two populations recur and must not be confused, which is the single most common way to misread a table like this. The *archived* population is the 500 instances generated under master seed 20260628. The *pooled* population is the $10 \times 500 = 5{,}000$ instances of the ten-seed campaign, of which the archived seed is one. Table 3 records, for every experiment, which population it uses and under what protocol.

Three quantities recur and are not interchangeable. The *flip rate* of a rule at level p is the fraction of bound draws at which that rule's winner differs from its own nominal winner, a within-rule measure that every rule has. *Winner divergence* is the fraction of instances at which two rules pick different alternatives at the nominal bounds, with no perturbation involved. *Modal frequency* is the share of draws won by the most frequent winner, and means nothing except relative to the declared sampling distribution.

The contract reports ties as a class, so the class is what a flip rate should really be measured on. A *point flip* records any change in the single alternative selected after tie-breaking, whereas a *class flip* requires the set of tied winners to change. The point flip rate measures the stability of the rule plus its tie-breaking mechanism, while the class flip rate isolates the stability of the rule itself. Both are measured.

Finally, the magnitude of these flip rates is inevitably tied to the specific multiplicative perturbation model. To confirm that the relative stability gaps are structural rather than an artifact of that choice, the decomposition was replicated under an alternative, scale-coherent model where perturbations are drawn proportionally to the active range of each criterion. The numerical rates change—at the 5% level, the raw baseline flips in 38.6% of draws while LexPR flips in 18.4%—but the gaps separating the proposed method from the scalarised baselines are fully preserved.

Table 3: Protocol of the sampled and certified experiments. The certified cost study and the supplier removal diagnostic are deterministic and have no sampling protocol to record. *Draws* is the number of bound draws per uncertainty level, written as a box budget for the certified row, which samples nothing. Every comparison uses the exact order at $\tau = 0$, ties reported as a class rather than broken, except the polyhedral study, which applies a declared secondary rule because its optimal set can be positive-dimensional. Each row is produced by one script in the archive.

Experiment	Inst.	Sds	Draws	Levels
Selection, divergence, ablation	500	1	—	—
Point fragility	500	1	60/lvl	$5, 10, 20\%$
Seed sensitivity	5,000	10	60/lvl	$5, 10, 20\%$
Decomposition	5,000	10	60/lvl	$5, 10, 20\%$
Margin quintiles	500	1	40	20%
Supplier flip curve	1	1	300/lvl	0–50%
Supplier interval	1	1	500/lvl	5–20%
Certified supp.	1	—	40,000 boxes	2–20%
Known-truth	11	1	none	$2, 5, 10\%$
Polyhedral study	50	1	—	—

Table 4: Decomposition of the fragility gap over the pooled population, 60 bound draws per level. $\mathrm{ASF}_{\mathrm{raw}}$ acts on the raw normalised matrix. Every other rule acts on re-anchored disappointments and differs only in the family and in how the profile is collapsed. The tail-loss column is defined where it is first discussed, in the inference paragraph below.

Rule	Flip rate			Tail	Agrees
	5%	10%	20%	loss	w/ LexPR
$\mathrm{ASF}_{\mathrm{raw}}$	36.4%	55.5%	75.1%	0.584	—
LexPR	18.3%	29.9%	44.8%	0.597	—
ASF_K	17.9%	29.5%	44.5%	0.597	98.0%
MMR_K	18.4%	30.0%	44.8%	0.597	97.6%
LexPR_1	0.0%	0.0%	0.0%	0.584	—
ASF_1	0.0%	0.0%	0.0%	0.584	—

5.1 Where the stability comes from

A naive comparison of LexPR against a raw-normalised achievement scalarisation is not calibrated: LexPR gets K probes and the baseline gets m equally weighted criteria, so any gap confounds the richness of the family, the re-anchoring, and the lexicographic order. Table 4 separates the three by running six rules on the same 5,000 pooled instances, each differing from its neighbour in one respect only. Read from the bottom up, the table tells a single story in four steps.

First, single-criterion probes are exactly bound-invariant. This is Proposition 1, not an empirical finding, and the measurement confirms it: the singleton-only rule flips in 0.0% of draws at every level to the displayed precision. The measured rate is not identically zero but one to two flips in ten thousand, against nineteen to twenty for the singleton-only achievement scalarisation. Those residual flips are tie-break events, not bound sensitivity: the identity removes the bounds from the disappointment but does not break ties, and when two alternatives have equal profiles the reported winner is decided by declared index order.

Second, adding aggregate probes is what introduces bound sensitivity. The mean and the maximum mix criteria whose ranges move at different rates, so they do not cancel, and declaring them is exactly what raises the flip rate from zero to 44.6% at a 20% perturbation. Richer declarations buy discriminating power and pay for it in bound stability. That is a lever a practitioner can pull, and it is all-or-nothing: a deployment more afraid of bound error than of coarse discrimination can declare a singleton-only family and get exact invariance, but one active aggregate probe loses it.

Third, the re-anchoring carries the whole gap against the baseline. Moving from $\mathrm{ASF}_{\mathrm{raw}}$ to a re-anchored rule takes the flip rate from 75.3% to about 44.5%. Measured against ASF_K, which differs from $\mathrm{ASF}_{\mathrm{raw}}$ in acting on re-anchored probe disappointments rather than on raw normalised criteria, the paired per-instance difference is $+0.310$, with instance-level interval $[+0.301, +0.319]$ and family-clustered interval

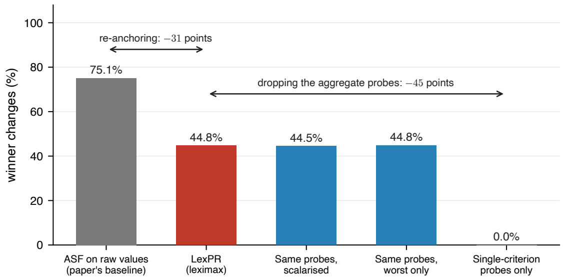

Fig. 6: Where the stability comes from. Re-anchoring accounts for the whole gap against the baseline, replacing the leximax by a scalarisation changes nothing, and a family of single-criterion probes never changes its winner at all, because those disappointments do not depend on the bounds. Pooled population.

$[+0.135, +0.458]$. This is a property of the disappointment construction of (2), shared by every rule built on it.

Fourth, the ordering itself does not move the needle. Collapsing the same profile with a Chebyshev scalarisation augmented by a small multiple of the mean, the rule listed as ASF_K, gives 44.4% against LexPR's 44.6%. The paired difference is -0.0024, with instance-level interval $[-0.0036, -0.0013]$ and seed-clustered interval $[-0.0035, -0.0013]$, both excluding zero, against a family-clustered interval $[-0.0063, +0.0002]$ that contains it. The sign test over the ten seeds gives $p = 0.021$. The effect is consistent in sign, negligible in size, and runs against LexPR rather than for it. Truncating the same profile to its worst coordinate alone, the rule listed as MMR_K, gives 44.8%, a difference of $+0.0016$, two orders of magnitude smaller than the effect it is compared against. Each of the two agrees with LexPR on the selected alternative in about 98% of instances. A word on resolution, because three rates within half a point of each other invite over-reading. With three hundred thousand draws per level the standard error of a single rate is about 0.09 percentage points. Only the *paired* per-instance differences, which cancel the shared draws, resolve gaps this small. Whatever the lexicographic refinement buys, and we argue in Section 2 that it buys the decisive coordinate and its margin, it is not point stability.

Figure 6 is the same four steps in one picture.

The honest restatement is therefore not "LexPR is more stable than ASF" but "rules built on active-range normalised probe disappointments are more stable than rules built on raw normalised criteria". That is a claim about the modelling object, not about the ordering placed on top of it, and none of it weakens the certified machinery, which reports which alternatives can win rather than asserting that one is stable.

5.2 Comparison with other rules, and how much it varies

Only now, with the decomposition settled, does a comparison against rules built on a different object mean anything. LexPR runs against achievement scalarisation, sampled minimax regret [25, 26], TOPSIS, compromise programming, VIKOR [27] and a rule that picks the single alternative of largest hypervolume contribution, all with equal weights while LexPR uses a richer declared family, so these are descriptive comparisons and not a calibrated contest.

LexPR and the achievement scalarisation select different winners in 66.2% of archived instances and 68.6% of pooled ones. Under the declared perturbation their point winners change in 43.8% and 75.2% of draws at 20%, and in 16.7% against 36.1% at 5% and 28.6% against 55.5% at 10%. Both point selectors are fragile. The interval mode does not lower the flip rate, it exposes the alternatives among which the flips occur.

An aggregate rate hides almost everything here, which Figure 7 makes plain. Divergence is near total on knapsack instances and near zero on WFG2. The fragility advantage is carried by three families, with paired differences of $[0.560, 0.595]$ on WFG2, $[0.384, 0.415]$ on DTLZ and $[0.366, 0.394]$ on knapsack, while on job-shop instances it collapses to $[0.028, 0.062]$ and on the two surrogate matrices sits around 0.09 to 0.15. WFG2 is the extreme case in both directions: the two rules agree on the winner in 90.5% of instances there, yet LexPR's flip rate is 3.6% against 61.4%, meaning they mostly pick the same alternative but only one of them keeps picking it as the bounds move. Any statement of the form "LexPR is less fragile" has to be qualified by generator family.

One measurement supports the positive claim we do make about the ordering, namely that it explains. For a unique winner and a rival, the *deciding coordinate* is the first sorted position at which they differ, and the worst-case explanation

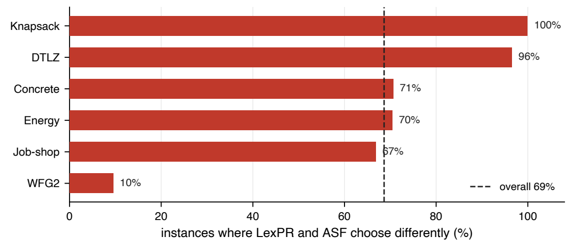

Fig. 7: How often the two rules pick different alternatives, by generator, on the pooled population. The dashed line is the overall rate of 68.6%, rounded to 69% in the label, and it is not the 66.2% of the archived population quoted in the text. The per-generator spread here is the same phenomenon as the spread in the fragility gap discussed below: an aggregate rate describes no single generator well.

depth of a winner is the largest such position over all rivals. Across the archived population, of which 498 of 500 instances had a unique winner, the mean worst-case depth is 1.04 coordinates with a standard deviation of 0.22. Almost always the winner's largest disappointment is already below every rival's, so the comparison is settled at the first coordinate and nothing deeper has to be shown to anyone. Because sorting discards the labels before comparing, this does not attribute the decision to one fixed declared question, and the figure is not claimed invariant to the criterion count or the shortlist size.

The declared family matters at least as much as the rule. Removing the single-criterion probes changes the winner in 90% of instances, collapsing to the mean alone in 97%, to the maximum alone in 68%, and a one-probe family equivalent to the achievement scalarisation diverges in 66%. The recommendation is a function of what was declared, and the declaration is the part a facilitator controls.

How much of this is sampling noise

Instances are nested inside master seeds and inside generator families, and the previous subsection shows the effect varying by more than a factor of ten across families, so an instance-level bootstrap is anticonservative here rather than merely conceivably so. We therefore recompute every headline difference under three resampling schemes with ten thousand resamples each: the instance-level scheme, a cluster bootstrap over the ten master seeds, and a hierarchical bootstrap that resamples generator families, then seeds inside family, then instances inside cell. We report the instance-level and hierarchical intervals together and read the hierarchical one as the pessimistic end of a range, since it has only six top-level units. An exact two-sided sign test on the ten per-seed differences adds a check that assumes nothing beyond exchangeability of seeds.

The fragility gap survives all three. The paired difference in flip rate is $+0.184$ with instance interval $[0.174, 0.193]$ and hierarchical $[0.021, 0.316]$ at 5%. At 10% it is $+0.258$, $[0.248, 0.268]$ and $[0.073, 0.407]$, and at 20% it is $+0.304$, $[0.295, 0.313]$ and $[0.130, 0.454]$. The hierarchical intervals are roughly sixteen to eighteen times wider and still exclude zero, and the sign test gives $p = 0.002$ at every level, the smallest value ten seeds can produce.

One result does not survive, and we report it in full because it is the one that matters. Each rule is scored on a pool of held-out synthetic preference families: ten declared utility families, each sampled with four hundred draws, none of them used to choose the probes. The *tail loss* of an alternative is the mean of the pooled losses at or above their empirical ninetieth percentile, so it measures how badly a choice can turn out rather than how it does on average. On that measure the achievement scalarisation wins on the point estimate, with a paired difference of -0.013. Here the three schemes disagree, and pretending otherwise would be dishonest. The instance-level interval is $[-0.014, -0.012]$ and the seed-clustered interval $[-0.0136, -0.0122]$, both excluding zero, and the sign test rejects at $p = 0.002$ with all ten seeds favouring the baseline. The family-clustered interval is $[-0.036, +0.007]$ and contains zero. What separates the schemes is between-family variation, so the defensible reading is that the disadvantage is consistent across seeds and not separable from differences between generators. We therefore do not claim the gap is established, and we do not claim it is absent either. What remains defensible is weaker and still uncomfortable: on these generators LexPR's point recommendation is measurably less fragile under bound perturbation and shows no held-out advantage, with the sign running against it but not separable from between-family variation. Reporting the instance-level interval alone would have made that difference look established, which is precisely the failure mode a clustered analysis exists to catch. The held-out families are author-designed and structurally overlapping, so this exercise is an out-of-rule scenario evaluation and not external validation.

Seed sensitivity is worth one sentence of its own, because it changes how the single-seed numbers should be read. The archived seed sits at the favourable edge for several quantities: LexPR's flip rate at 5% is 16.7% on it against a pooled 18.0%, and that is the seed minimum, as is the divergence rate of 66.2% against a pooled 68.6%. Between-seed standard deviations run from 0.7 to 1.3 percentage points, so no qualitative conclusion changes, but the single-seed figures are optimistic by one to two points.

5.3 Knowing in advance what kind of answer to expect

LexPR often returns a class rather than a point, which is honest but awkward to plan around. The contrastive margin $\Delta_\mathcal{C}$ of Section 2 makes that predictable before any perturbation study is run, at no extra cost, because the finite algorithm already reports it. Over the archived population its Spearman correlation with the flip rate at 20% is -0.39, ahead of the criterion count at -0.29 and far ahead of the candidate-set size, which contributes essentially nothing. It is not the strongest predictor available: the realised maximum correlation between criteria reaches -0.43 on the same data. We nonetheless report the margin, because it is produced by the selection itself at no extra cost and is interpretable as the closeness of the decision, whereas the correlation is a property of the instance that says nothing about which alternative won.

Grouping by quintile makes it usable (Figure 8). In the bottom quintile, $\Delta_\mathcal{C} \leq 0.005$, instances flip in 58.5% of draws and yield a singleton class only 12% of the time, with a mean class of six alternatives. In the top quintile, $\Delta_\mathcal{C} > 0.031$, the flip rate falls to 26.4% and singletons rise to 34%. A facilitator can read the margin off the first run: below roughly 0.01 on the normalised scale, the defensible output is a class. Two caveats keep this a screening device

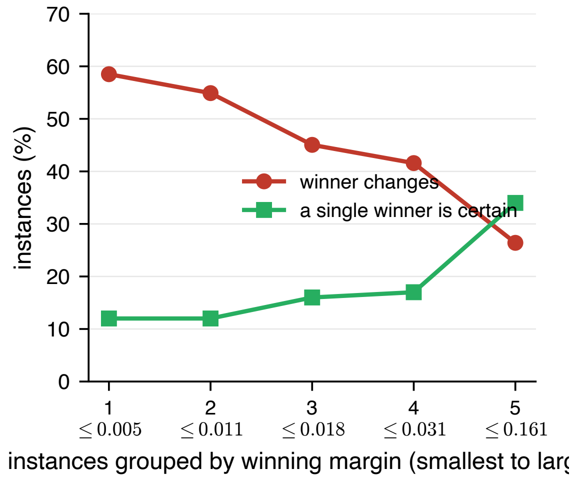

Fig. 8: The winning margin predicts what kind of answer to expect. Instances where the winner beats its closest rival by little change winner often and rarely admit a single defensible choice. Archived population.

rather than a substitute for the interval analysis. The relationship is monotone but far from deterministic, and even the top quintile produces a non-singleton class in two thirds of instances.

5.4 One instance, end to end

A constructed supplier-selection instance exercises the whole contract on a problem small enough to inspect. **The data are constructed, not observed**: the matrix was designed to contain conflicting criteria, one near-collinear environmental pair and ten efficient alternatives. It carries no external validity and is an illustration, not a case study.

A buyer must choose one of 10 pre-screened suppliers scored on 7 cost criteria: price, quality, lead time, two environmental measures, social risk and supply resilience [28, 29]. The two environmental criteria are near-collinear, correlation 0.993. All ten alternatives are efficient, so this is a genuine selection. The declared family is the canonical core, $K = 9$.

Ten rules split across four suppliers. LexPR returns S7, whose profile is Figure 3: worst on defect rate (0.79), price (0.69) and resilience risk (0.53), essentially zero on both environmental criteria, low on social risk and lead time. That is the whole explanation, and it is readable by someone who has not read this paper.

Then the bounds are perturbed. The point recommendation is fragile, changing in 52% of draws at 20%, but the interval mode narrows the answer rather than dissolving it (Figure 9): the winner union is $\{\text{S1}, S7\}$ even at 20%, with frequencies 51% and 49%. Those frequencies carry Monte Carlo uncertainty and must be reported with it: over 500 draws the Wilson score intervals are $[46.6\%, 55.4\%]$ and $[44.6\%, 53.4\%]$, which overlap and both contain 50%. The deliverable is therefore not "choose S7" but "the winners are S7 and S1, at these frequencies under the declared distribution", with both profiles attached.

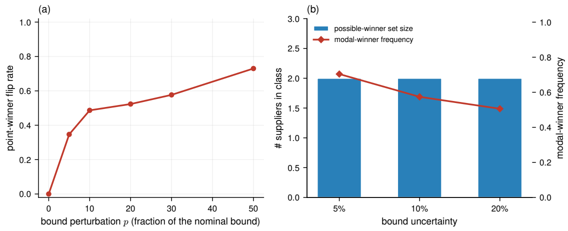

Fig. 9: The point recommendation is fragile and the interval mode says so. Left, the winner changes more often as the bounds are perturbed. Right, that fragility becomes a small named class: two suppliers at every level, with the modal share falling towards a coin flip as the box widens.

The certified analysis sharpens this into a guarantee. At a 2% box the procedure eliminates eight of the ten suppliers outright, and membership of the remaining two is established by explicit witnesses, so the exact possible set is $\{\text{S1}, S7\}$ and the necessary set is exactly empty. The certification frontier for S7 is 0.23% and a witnessed reversal to S1 occurs at 1.3%, so by (7) $\rho^\star \in [0.23\%, 1.3\%]$. Both witness vectors are printed in the archive, so the conclusion can be re-checked without rerunning the search.

Two diagnostics complete the picture. Removing one non-selected supplier at a time changes the recommendation once in nine, a rate of 11%, and LexPR is *not* more robust here than TOPSIS, compromise programming or minimax regret, all at the same rate. Only the achievement scalarisation is worse, at 22%. Candidate-set dependence belongs to the normalisation, not to one rule, exactly as Theorem 1 says. And removing the two optional cluster probes leaves the winner and its profile unchanged, so this instance, the most favourable available test of that layer since it contains a genuine correlated pair, gives no evidence that it improves anything.

This gives the deployment rule. When the shortlist, bounds, anchors and family are frozen and the margin conditions certify a unique winner, report the point and its profile. Otherwise report the stability class, reading the modal winner only under the declared distribution and declining a unique recommendation when the class is large. For many real shortlists the output will be a class, because the stability radius of a point winner can be very small: here the bracket is only $[0.23\%, 1.3\%]$ of the nominal bounds, tighter than most estimation processes deliver.

6 Scope, limitations, and what to take away

LexPR is useful when the decision maker cannot commit to precise trade-off weights but can declare and defend a finite family of monotone scalar questions, and when the recommendation has to be explained and re-checked afterwards. It is not the right tool when the objective is average-case utility under a known preference distribution, when no meaningful monotone probes can be declared or the family is itself the political dispute, or when the bounds are so uncertain that the stability class stays large. In that last case the method still gives the right answer, which is that the data do not support naming one alternative.

Three limitations are structural rather than fixable by more computation. The rule depends on the candidate set, because active-set normalisation ties the scale to the observed extremes, and Theorem 2 says exactly which edits are safe. Robustness holds relative to the declared family rather than over all conceivable preferences, as the probe ablation shows. And the construction protects against the worst case rather than maximising average utility, which the held-out results reflect.

Several threats to validity deserve naming. The supplier case is reproducible but constructed, so validation on real decision problems is the most important next step. The held-out preference pool is author-chosen, weights ten families equally, and contains two that share a power-mean form, so that geometry counts twice. The certified mode reaches only small boxes and modest criterion counts. And the polyhedral formulation is proved for canonical probes over polyhedra with affine criteria, with the mixed-integer branch untested and general convex probes out of scope.

The takeaway is not a new decision rule. The ordering is classical and we claim nothing for it. It is that the last step of a multicriteria study, turning an efficient set into a defensible choice, can be treated as a computational problem, with the guarantees that implies. An exact answer on the shortlist. A certified answer over a continuum of admissible normalisations rather than a sampled one. A formulation that skips enumeration. And a record from which someone else can rebuild the conclusion. The empirical work then locates the source of its own effect and finds it is not where the name of the method suggests. The stability comes from normalising each declared question on the range that question actually spans. The lexicographic comparison earns its place by explaining the choice, worst concern first, in the terms the problem was posed in.

Declarations

Competing interests. The author declares no competing interests.
Funding. No external funding supported this work.
Ethics approval. Not applicable. The study uses no human participants, no animal subjects and no personal data.
Data and code availability. All data, code and seeds are in the reproduction archive cited below. The supplier instance is constructed and is distributed with the archive. No third-party or proprietary data is used.
Relationship to a previous preprint. An earlier and longer version of this work was deposited as a preprint (Zenodo record 21037960). The present paper supersedes it. Two classes of claim made there are withdrawn rather than restated: an "auditability trilemma" framing, and the stronger axiomatic reading of the profile-order result, which is a representation lemma relative to a declared family and not a derivation of the leximax order from primitive postulates. The empirical claims have also been narrowed, as Table 1 records.

Use of large language models. A large language model was used as a writing aid: to shorten prose, to check the reported numbers against the archived outputs, and to draft regression tests. It was not used to generate results or to design experiments. Every quantity here is computed by the archived scripts, and the author is responsible for the whole content.

Reproducibility

All code, data, seeds and scripts are in the archive [19], version 1.0.4, at https://doi.org/10.5281/zenodo.21739625 and tagged v1.0.4 at https://github.com/MadBezoui/LexPR. One command, `python scripts/reproduce_all.py`, regenerates every number and figure here from the fixed master seeds. The archive also holds the full proofs, the algorithm listings with their enclosure formulas, the per-row output of Table 2, the two supplier witness vectors and the audit records.

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